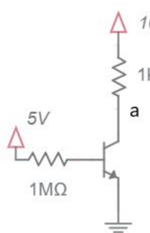
ELECTRICAL ENGINEERING DEPARTMENT
COLORADO SCHOOL OF MINES

DC circuits with NPN BJTs and resistors

Determine the voltage at the labeled nodes. Determine collector, base and emitter currents.

- If the BJT is in the active region, assume V_{be} = 0.7V, ß = 100, and i_c=i_e.
- If the BJT is in saturation, assume that V_{ce} = 0.2V.
- Only compute ß when the BJT is in saturation.
- Write your answers on the table shown on the page following the problems.
- On separate page(s), show your work for each problem. Denote the operating region of BJTs by writing ACT/SAT next to active/saturated mode BJTs respectively.
- Write “unknown” for circuits whose current/voltage cannot be determined.
- Round you answer to 2-significant figures and include units.



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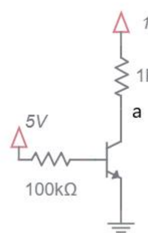

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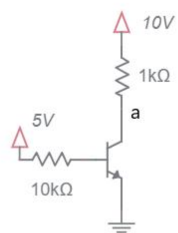

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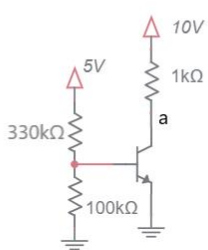

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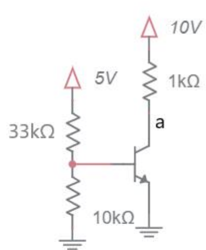

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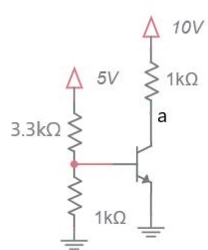

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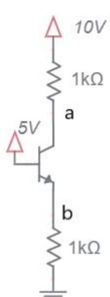

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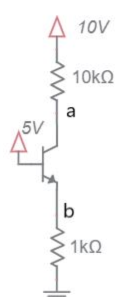

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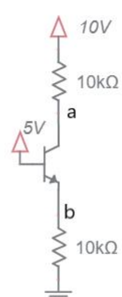

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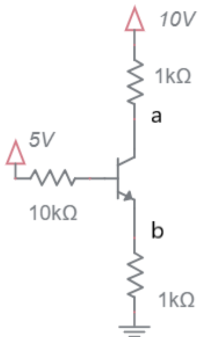

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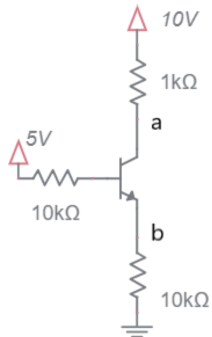

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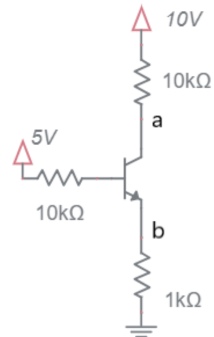

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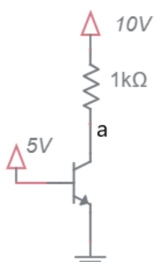

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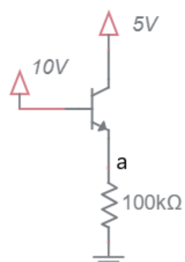

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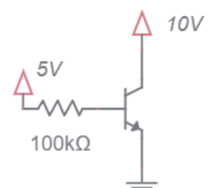

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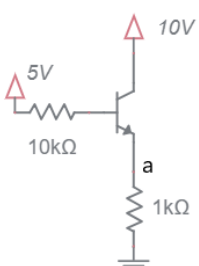

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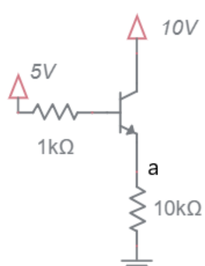


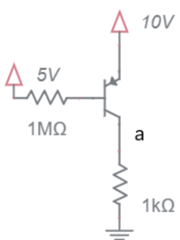
017

Problem		a	b	ic	ib	ie	ß
001		9.6V		0.43mA	4.3uA	0.43mA	
002		5.7V		4.3mA	43uA	4.3mA	
003		0.2V		9.8mA	0.43mA	10.23mA	23
004		9.4V		0.6mA	6uA	0.6mA	
005		4V		6mA	60uA		
006		0.2V		9.8mA	0.6mA	10.4mA	16
007		5.7V	4.3V	4.3mA	43uA	4.3mA	
008		4.5V	4.3V	0.55mA	3.8mA	4.3mA	0.73
009		5.7V	4.3V	0.43mA	4.3uA	0.43mA	
010		6.1V	3.9V	3.9mA	39uA	3.9mA	
011		9.6V	4.3V	0.43mA	4.3uA	0.43mA	
012		1.3V	1.1V	0.87mA	0.32mA	1.1mA	2.7
013		9.3V			unknown		
014				unknown			
015				0.43mA	4.3uA	0.43mA	
016		3.9V		3.9mA	39uA	3.9mA	
017		4.3V		0.43mA	4.3uA	0.43mA	

DC circuits with PNP BJTs and resistors

Determine the voltage at the labeled nodes. Determine collector, base and emitter currents.

- If the BJT is in the active region, assume V_{eb} = 0.7V, ß = 100, and i_c=i_e.
- If the BJT is in saturation, assume that V_{ec} = 0.2V.
- Only compute ß when the BJT is in saturation.
- Write your answers on the table shown on the page following the problems.
- On separate page(s), show your work for each problem. Denote the operating region of BJTs by writing ACT/SAT next to active/saturated mode BJTs respectively.
- Write “unknown” for circuits whose current/voltage cannot be determined.
- Round you answer to 2-significant figures and include units.



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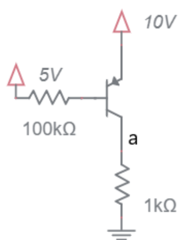

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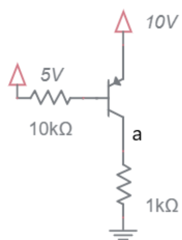

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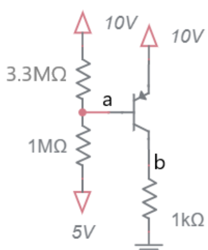

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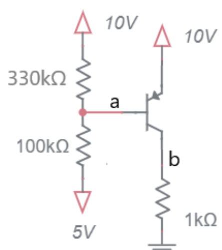

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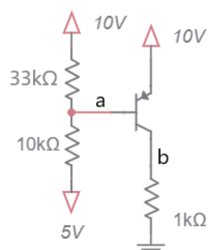

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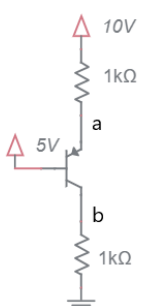

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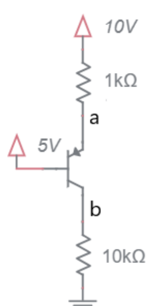

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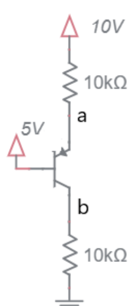

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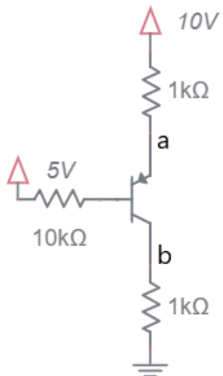

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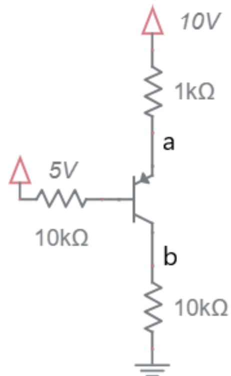

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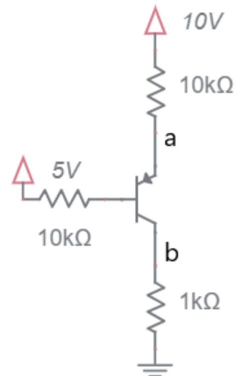

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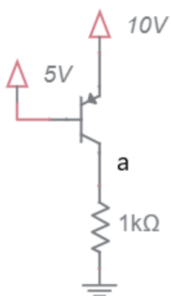

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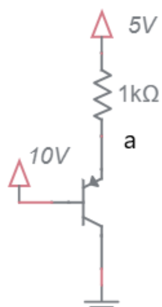

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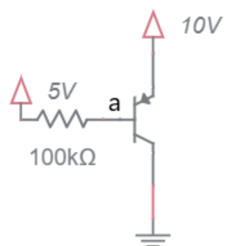

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002		4.3V		4.3mA	43uA	4.3mA	
003		9.8V		9.8mA	0.43mA	10.2mA	23
004		9.3V	0.41V	0.41mA	4.1uA	0.41mA	
005		9.3V	4.1V	4.1mA	41uA	4.1mA	
006		9.3V	9.8V	9.8mA	0.41mA	10.2mA	24
007		5.7V	4.3V	4.3mA	43uA	4.3mA	
008		5.7V	5.5V	0.55mA	3.8mA	4.3mA	0.15
009		5.7V	5.3V	0.53mA	5.3uA	0.53mA	
010		6.1V	3.9V	3.9mA	39uA	3.9mA	
011		6.3V	6.1V	0.61mA	3.1mA	3.7mA	0.2
012		5.7V	0.43V	0.43mA	4.3uA	0.43mA	
013				unknown			
014		9.3V				unknown	
015		9.3V		4.3mA	43uA		